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Paleosecular variation record of geomagnetic full vector during late Miocene, from the Nayarit area, Mexico

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Abstract

We have sampled a sequence of 45 late Miocene consecutive lava flows in the Tepic area (Nayarit State, Mexico). The age of the volcanic units lies between 8 and 9 million years (Ma) according to available radiometric data. All samples were stepwise demagnetized, partly with alternating field (AF), partly thermally with very similar results. Most of the rocks exhibited well-defined one component remanent magnetisation with high unblocking temperatures (mostly above 525 °C) and high median destructive fields (MDF) (40–50 mT). Rock-magnetic experiments combined with microscopy show that, in most cases, the main magnetic mineral is Ti-poor titanomagnetite associated with exsoluted ilmenite. Continuous susceptibility measurements with temperature and hysteresis experiments yield in most cases nearly reversible curves with Curie points close to that of magnetite and pseudo-single-domain characteristics. Characteristic remanent magnetisations (ChRM) isolated after the first steps of demagnetisation are all normal polarity. According to the dispersion of virtual geomagnetic pole (VGP) directions, paleosecular variation was abnormally lower than the one observed in general during Miocene. Considering our paleomagnetic results together with available radiometric data, it seems that the volcanic units have been emplaced during a very short time span of about 0.08 million years. The mean paleomagnetic directions obtained from this study do not differ significantly from that expected for the middle Miocene. Thirty-one samples from eight individual flows yielded acceptable paleointensity estimates. The site mean paleointensities range from 27.8 ± 0.9 to 42.0 ± 7.9 μT. The virtual dipole moments (VDM) range from 5.9 to 9.5×10^{22} Am^2. This corresponds to a mean value of $7.6 \pm 1.4 \times 10^{22}$ Am^2, which is higher than the average VDM value for late Miocene. Altogether our data suggest the existence of relatively high geomagnetic field strength undergoing low fluctuations. These results support the theoretical suggestion about an inverse relationship between secular variation and local field strength as result of electromagnetic coupling between the solid inner core and liquid outer core, with the inner core tending to stabilise core convection, and hence the field, when intensity is high. Some fluctuation of absolute intensity was detected within the same directional group (DG) indicating that the intensity of the geomagnetic field varies faster than its direction.
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1. Introduction

Knowledge of the variation of Earth's magnetic field with geological time helps to understand how the

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geodynamo works. These data provide thus an important source of information on the physics of the Earth's deep interior. Records of geomagnetic paleosecular variations may be obtained using either sedimentary or volcanic rocks. Sediments can provide continuous records of magnetic field variation, while lava data, due to the sporadic nature of volcanic activity, yield rather discontinuous records of geomagnetic field variation. On the other hand, the results obtained from lavas are not subject to the controversies over reliability that make interpretation of sedimentary data so difficult (Tauxe, 1993; Dunlop and Özdemir, 1997; Love, 2000). Moreover, volcanic data can yield, in some cases, absolute intensities, while sedimentary data can only give relative intensity variation. Camps and Prévot (1996) and more recently Love (2000) showed that in any correct model of the geomagnetic field, the absolute intensity should necessarily be incorporated. Moreover, Coe et al. (2000) suggested that paleointensity of the Earth magnetic field should be a fundamental constraint in numerical models that promise to provide unprecedented insight into the operation of the geodynamo. However, the studies that involve both paleodirections and paleointensities are extremely scarce (Love, 2000) and cannot be yet used to document the general characteristics of the Earth's magnetic field.

Paleomagnetic data in volcanic units provide spot-readings of the evolution of the past geomagnetic field. Studies of the statistical dispersion of the paleomagnetic directions is the only way to describe long-term variations. Such studies of the paleosecular variations are numerous for the last 5 million years (Ma). Few data are however available for the Miocene period (McEllhinny and McFadden, 1997) and their geographic distribution is still very uneven (e.g. Tanaka et al., 1995; Perrin and Shcherbakov, 1997). In particular very few data are available from central and South America, which impedes an accurate analyses of the fine-scale changes in the statistical characteristics of paleosecular variations (Jacobs, 1994). Up to now, there are very few studies reporting paleomagnetic results in volcanic rocks for the Latin America (Goguitchaichvili et al., 2000a). Most studies belong to the Andean chain and it is thus difficult to separate local tectonic rotation from true paleosecular variation

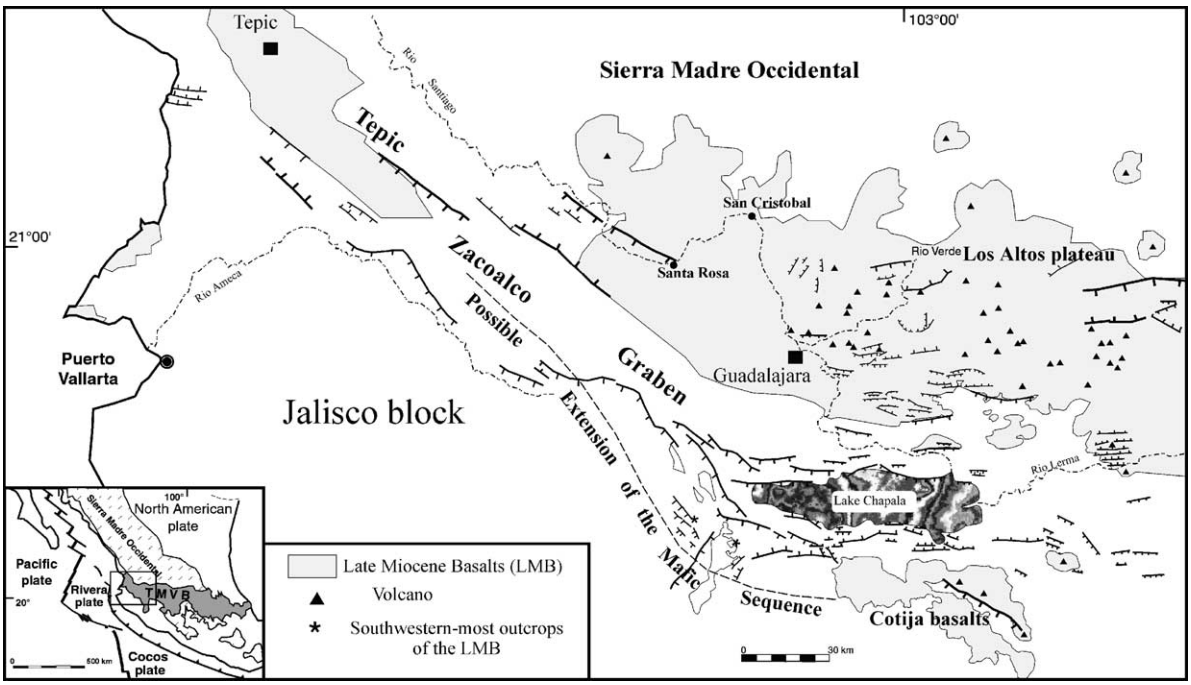


Fig. 1. Simplified geologic map of late Miocene mafic lavas and main fault systems of the western Trans-Mexican volcanic belt.

(Roperch et al., 1996). The main purpose of the present work is to make an effort for improving our knowledge of the past behaviour of the Earth's magnetic field in terms of both directions and intensity from Mexico during late Miocene. These data also provide better constraints on the age, duration and extrusion rate of volcanic units. In the present work, we have studied in detail the single sequence from western Mexico constituted by 45 consecutive, independent lava flows and characterized by unambiguous stratigraphy, so that sampling artifacts due to imperfect correlation and possible overlaps are excluded.

2. Geological setting, ages and sampling details

As the East Pacific Rise approached the North American plate through time, the Farallon plate was broken into smaller plates (e.g. Rivera plate; Atwater, 1970). In consequence, the middle tertiary Sierra Madre occidental (SMO) and the late Miocene–Holocene trans-Mexican volcanic belt (TMVB) volcanic arcs were formed by tertiary to present-day subduction along the Pacific margin of Mexico (Fig. 1) coexisting now in space. The SMO is a wide silicic volcanic succession built from 32 to 20 million years (McDowell and Clabaugh, 1979). In the Tepic area this volcanic province ranges between 18 and 21 million years (Gastil et al., 1978). The southwestern sector of the SMO is affected by several north–northwest trending half-graben structures associated to 17–9 million years extensional tectonics with tilted blocks dipping to the north–northeast (Nieto-Samaniego et al., 1999).

The TMVB is a volcanic arc oriented roughly east–west formed during the last ∼10 million years in response to subduction of the Cocos and Rivera plates along the middle America trench. Two main distinctive

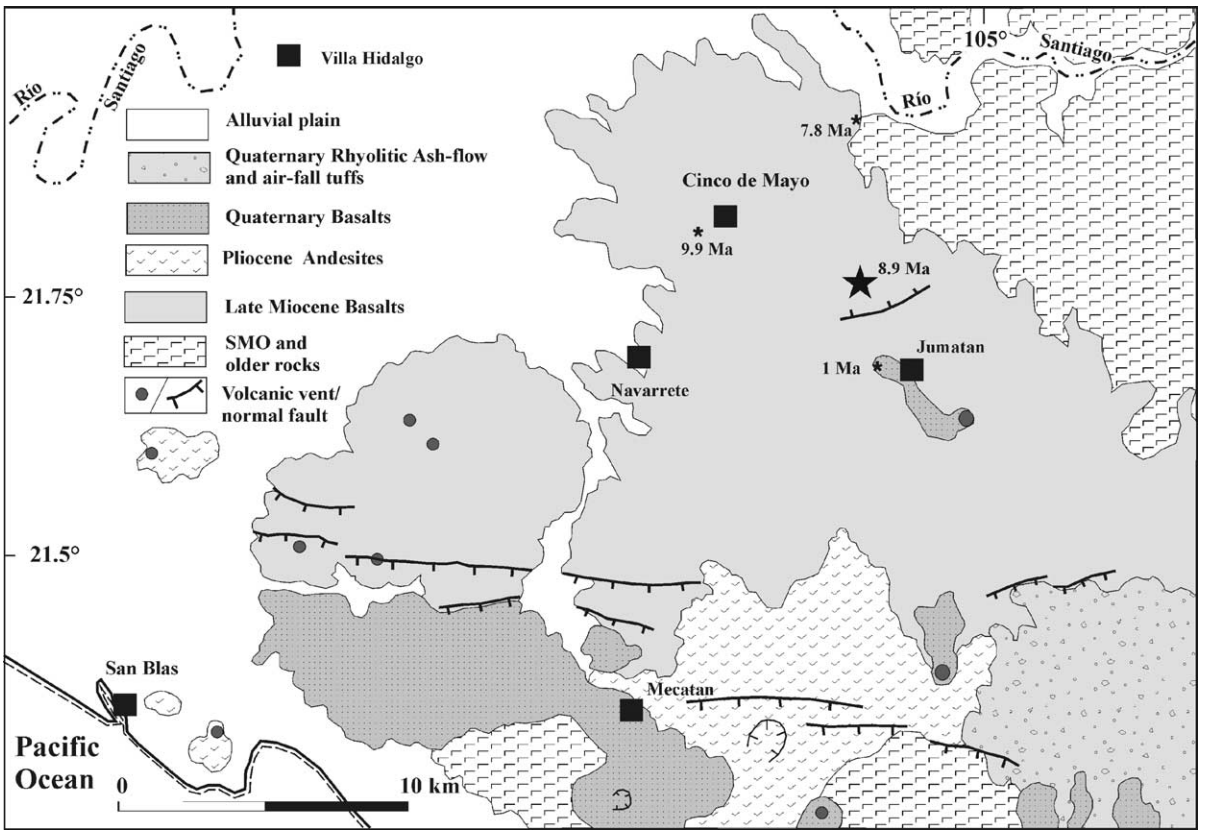


Fig. 2. Schematic geologic map of study area showing the location of volcanic sequence (★).

geological terrains can be recognized: a late Miocene basaltic unit mantled by the Plio-Quaternary volcanism; the first unit has a broad distribution and can be observed from the Tepic area to central Mexico along an east–west trending belt. This belt has only been documented to the north of the present-day TMVB. However, recent reconnaissance geological and paleomagnetic studies (Rosas-Elguera et al., 2002) demonstrated its extension towards the Cotija area (Fig. 1).

The opening of the Gulf of California and the formation of the contemporaneous widespread basalts at the western part of TMVB (Gastil et al., 1978) are the two most conspicuous events for late Miocene period. Because in the Tepic area no volcanic cones were detected we associate this volcanism to fissure lava flows which form plateau-like structures (Figs. 1 and 2).

There is still a controversy related to the geochemical affinity of the Tepic basalts. Earlier studies suggest an alkali signature (Righter et al., 1995). However, Ferrari et al. (2000) found a calc-alkaline character for the same samples used by Righter et al. (1995). In any case, there is no doubt that the studied lavas belong to late Miocene (Fig. 2). Southwest of Cinco de Mayo, Gastil et al. (1978) obtained a K–Ar age of 9.9 million years. From the same locality Righter et al. (1995) obtained 8.93 ± 0.11 million years using $^{40}Ar/^{39}Ar$ systematics. Compatible ages were reported for the west coast at Punta Raza, 67 km southwest of Tepic (Gastil et al., 1978). In southwestern Nayarit, Jensky (1975) reported a 10 million years-old basalt. Furthermore, the late Miocene basalts crop out as far as 105 km southwest of Tepic, in the Punta Mita area (Jensky, 1975).

The largest exposure of upper Miocene basalts was found east of San Blas (Fig. 2). This volcanic sequence is constituted by 45 consecutive lava flows. The whole sequence is affected by an echelon normal faulting related with the Mecatan graben (Ferrari and Rosas-Elguera, 2000). Although the general dip of this plateau-like structure is to the north, the sequence exposed along the highway 15 is tilted 15–20° southwest because of a small normal fault (Fig. 2). Righter et al. (1995) proposed, based on $^{40}Ar/^{39}Ar$ data, a mean age of 8.91 ± 0.06 million years as the best estimate of the time of emplacement of the volcanic sequence. The succession is formed by thin lava flows (<4 m), giving at least 200 m in thickness, without evidence of hiatus between them (e.g. baked sediments or paleo-soils). However, a well-defined contact zone may be clearly and systematically observed between each two consecutive lava flows.

In total, about 300 oriented samples belonging to 45 consecutive lava flows were collected. Commonly, the outcrops extend laterally over a few meters and

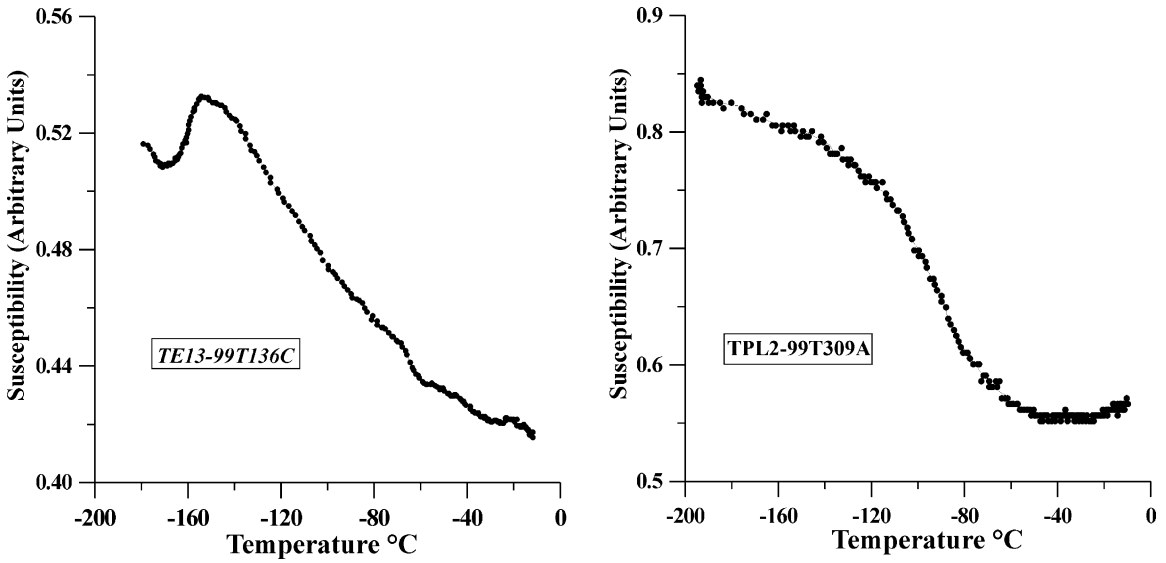


Fig. 3. Typical susceptibility vs. temperature curves recorded from about −185 °C to room temperature.

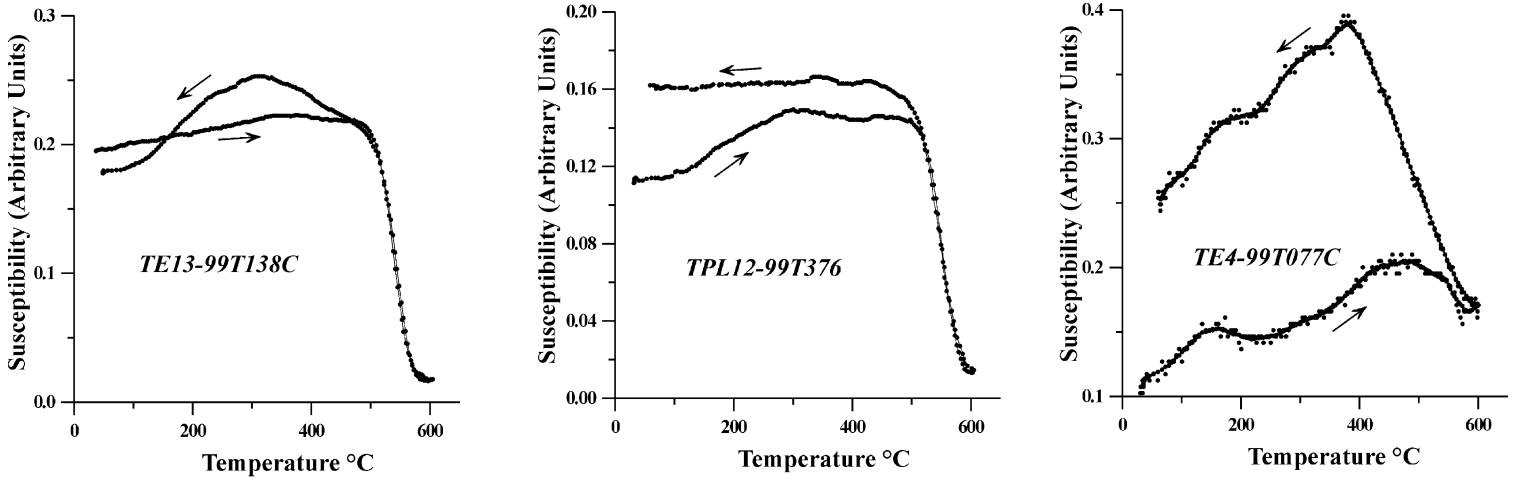


Fig. 4. Susceptibility vs. temperature curves recorded from room temperature to 600 °C. The arrows indicate the heating and cooling curves.

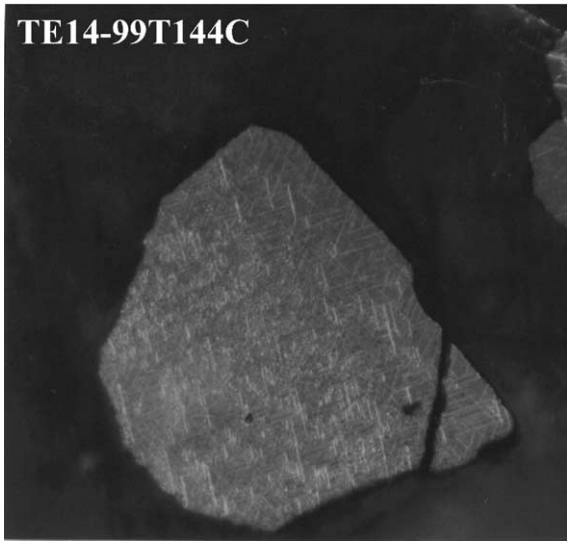


Fig. 5. Representative reflected light microphotographs from Tepic basalts, oil immersion, crossed nicols, dimensions 75 μm × 70 μm. Examples of ilmenite intergrowth of trellis type (Haggerty, 1976) in titanomagnetite transformed into Ti-poor titanomagnetite.

in these cases we drilled typically six to eight standard paleomagnetic cores per flow. The samples were both horizontally and vertically distributed throughout each flow in order to minimise effects of block tilting and lightning. In general, samples were obtained at the very bottom of flows with the hope of collecting samples with the finest grained material. At a few very small outcrops, only five samples were collected. Cores were obtained with a gasoline-powered portable drill, and then oriented with both magnetic and sun compasses.

3. Laboratory techniques

3.1. Viscosity index measurements

To obtain reliable and precise measurements of the remanence during progressive thermal demagnetisation it was recommended (Prévot et al., 1983) that magnetic viscosity be low. Xu et al. (1986) suggested that viscous remanences can be acquired very rapidly, but however can decay much more slowly, and that this process can operate in a wide variety of rock types. This phenomenon, perhaps due to a diffusion-after effect (Néel, 1955), could also impede accurate remanence measurements when the rocks are not treated and measured in a non-magnetic room. The viscosity index ν (Thellier and Thellier, 1944) was determined for pilot samples (two samples per flow) by measuring the remanent magnetisation first after a 2-week storage with the ambient field parallel to the cylindrical axis of each specimen, and then another 2-week

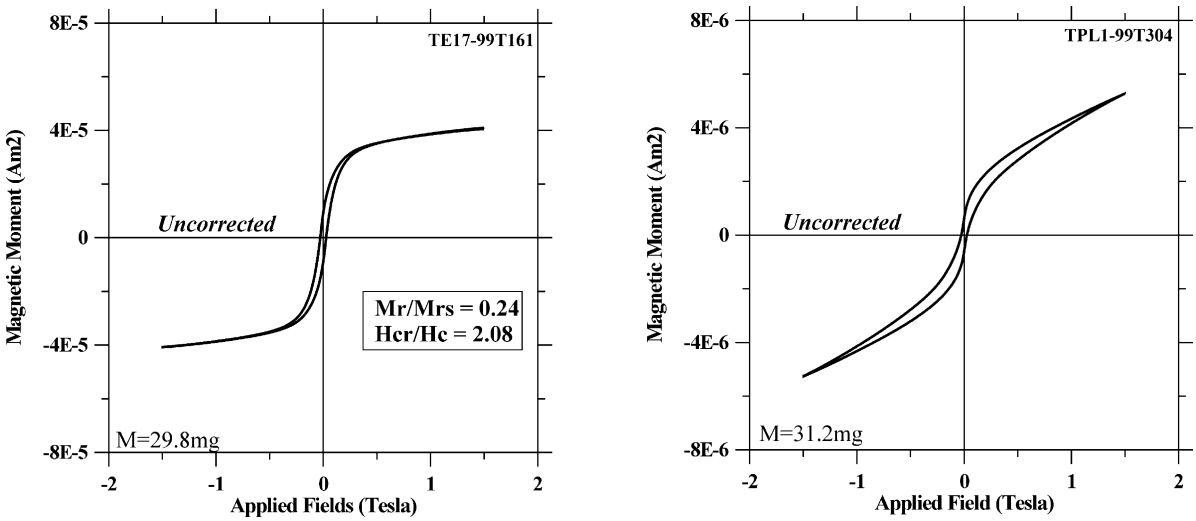


Fig. 6. Typical examples of hysteresis curves (uncorrected) of small chip samples from the studied volcanic units.

storage in the opposite direction. The studied samples show very low capacity of acquiring viscous remanent magnetisation. Although viscosity indexes varied between 0 and 6.8, most values are lower than 4%.

3.2. Low and high temperature continuous susceptibility curves

Low-field susceptibility measurements (*k–T* curves) under air were carried out using a Highmoore susceptibility bridge equipped with furnace. One sample from each site was heated up to about 600 °C at a heating rate of 20 °C/min and then cooled at the same rate. Curie temperature was determined by the Prévot et al.'s (1983) method. Alternatively, low-temperature (from about −185 °C to room temperature) susceptibility was recorded using the same apparatus.

Two different types of behaviour were observed during low-temperature susceptibility experiments (Fig. 3). In some cases, the curve shows a rather monotonic decrease from about −190 °C to room temperature (Fig. 3, sample 99T309A). In other cases a well-defined peak at about −145 °C was observed (Fig. 3, sample 99T136C), which is probably indicative to a Verwey transition, characteristic of almost pure magnetite. In both cases, some titanium-poor titanomagnetite may be responsible for remanent magnetisation. As showed by Özdemir and Dunlop (1993), the Verwey transition may be largely suppressed for titanomagnetites with variable titanium contents. Alternatively, similar behaviour can also be attributed to non-stoichiometric (partially oxidized) magnetite (Dunlop and Özdemir, 1997).

High-temperature susceptibility experiments also indicate, in most cases, the presence of Ti-poor titanomagnetites (Fig. 4, sample 99T138C). The cooling and heating curves are not perfectly reversible probably due to the low sensitivity of the Highmoore susceptibility system. However, these curves may be still used to determine the Curie temperatures of magnetic carriers. Microscopic observations on polished sections show that the main magnetic mineral is low-Ti titanomagnetite associated with exsoluted ilmenite (trellis or composite type after Haggerty's (1976) classification, Fig. 5) probably formed as a result of oxidation of titanomagnetite during the initial flow cooling. These intergrowths typically develop at temperatures higher than 600 °C (Haggerty, 1976)

Table 1
Site mean paleodirections of cleaned remanence and corresponding VGP positions for Tepic volcanics

Lava	DG	*n*/*N*	Inc°	Dec°	α_{95}	*k*	P_{lat}	P_{long}
TPU1	DG11	7/7	36.4	354.2	6.6	84	84.4	150.9
TPU2	DG11	6/7	33.5	2.2	5.5	196	86	43.2
TPU3	DG11	7/7	36.5	356.8	3.2	344	86.7	140.5
TPU4	DG11	7/7	40.4	354.2	4.1	234	84.5	180.2
TPU5	DG11	6/6	37.6	354.6	1.9	1428	84.9	158.7
TE1	DG11	6/6	38.9	1.1	2.7	98	88.9	329.1
TE2	DG10	6/6	44.1	359.3	3.1	126	85.8	246.4
TE3	DG10	6/6	44.7	355.8	3.8	257	83.9	216.2
TE4	DG10	6/6	44.4	1.2	4.5	219	85.5	268.8
TE5	DG09	7/7	39.5	359.7	5.7	182	89.2	233.4
TE6	DG09	7/7	34.6	5.7	4.5	292	84	10.5
TE7	DG09	8/8	37.4	359.8	3.7	126	89.2	88.5
TE8	DG08	6/6	27.7	3.8	4.4	225	82.1	47.1
TE9	DG08	6/6	32.6	1.3	4.8	261	85.8	57.6
TE10	DG08	6/6	35.3	0.1	2.6	153	87.8	72.6
TE11	DG07	5/6	31.9	359.2	2.3	158	85.5	84.8
TE12	DG06	6/6	37.1	356.5	3.3	554	86.6	148.8
TE13	DG06	7/7	37.8	0.2	2.1	845	89.5	54.6
TE14	DG06	5/6	34.1	355.5	8.1	235	84.8	130.4
TE15	DG06	7/7	35.1	353.1	4.5	133	83.1	146.3
TE16	DG06	6/6	34.3	353.3	3.9	297	83.1	141.7
TE17	DG06	6/6	35.1	350.9	4.1	269	81.2	151.3
TE18	DG06	6/6	40.4	354.8	3.4	502	85	181.7
TE19	DG05	6/6	33.5	355.3	2.8	177	84.4	128.3
TP_1	DG04	7/7	42.8	356.1	3.4	154	85.2	207.1
TP_2	DG04	6/6	36.2	357.1	4.3	228	86.9	134.9
TPL1	DG04	6/6	39.6	356.7	4.6	157	86.8	179.8
TPL2	DG04	7/7	38.5	358.2	2.6	539	88.3	164.9
TPL3	DG04	6/6	41.2	353.4	4.1	223	83.6	183.9
TPL4	DG04	7/7	40.5	352.6	2.5	583	83	178.2
TPL5	DG04	7/7	41.1	355.8	1.6	1506	85.7	191.5
TPL6	DG04	7/7	38.6	357.6	2.6	691	87.8	167
TPL7	DG04	6/7	34.8	357.8	6.8	79	86.7	114.5
TPL8	DG04	6/6	37.9	359.9	2.6	886	89.6	87.2
TPL9	DG04	7/7	37.1	0.6	3.2	359	88.9	45.3
TPL10	DG04	7/7	42.4	1.9	3.3	345	86.7	286.2
TPL11	DG03	6/7	34.2	356.6	5.5	197	85.7	123
TPL12	DG03	5/5	31.5	357.1	3.6	656	84.6	105.9
TPL13	DG03	6/6	30.3	1.8	1.7	2031	84.3	57.2
TPL14	DG02	5/6	36.4	355.1	4.6	275	85.2	148.1
TPL15	DG02	6/7	36.3	353.3	7.3	64	83.6	152.5
TPL16	DG02	6/6	38.4	353.4	5.8	252	83.9	165.5
TPL17	DG02	6/6	36.4	353.9	5.9	165	84.1	151.7
TPL18	DG02	6/6	36.2	350.2	1.2	3224	80.7	156.8
TPL19	DG01	7/7	29.2	358.5	3.8	252	83.7	88.4

N is the number of treated samples; *n* the number of specimens used for calculation; Dec the declination; Inc the inclination; *k* and α_{95} the precision parameter and radius of 95% confidence cone of Fisher statistics, respectively; and P_{lat}/P_{long} is the latitude/longitude of VGP position. Sites are listed in stratigraphic order.

and consequently, the natural remanent magnetisation (NRM) carried by these samples should be a thermoremanent magnetisation (TRM). It should be noted that the size of the magnetic grains observed is often too small and the petrological/optical observations of larger ferrimagnetic minerals may not completely correlate with the magnetic behaviour of the whole rock. In a few cases (Fig. 4, sample 99T077C) a well-marked irreversibility between cooling and heating curves was observed. This behaviour is probably due to a drastic magneto-chemical transformation which occurs during heatings in air. We suspect titanomaghemite to be the remanence carrier in this sample which dissociates at low/moderate temperatures producing pure magnetite (Readman and O'Reilly, 1972; Özdemir, 1987). This sample was

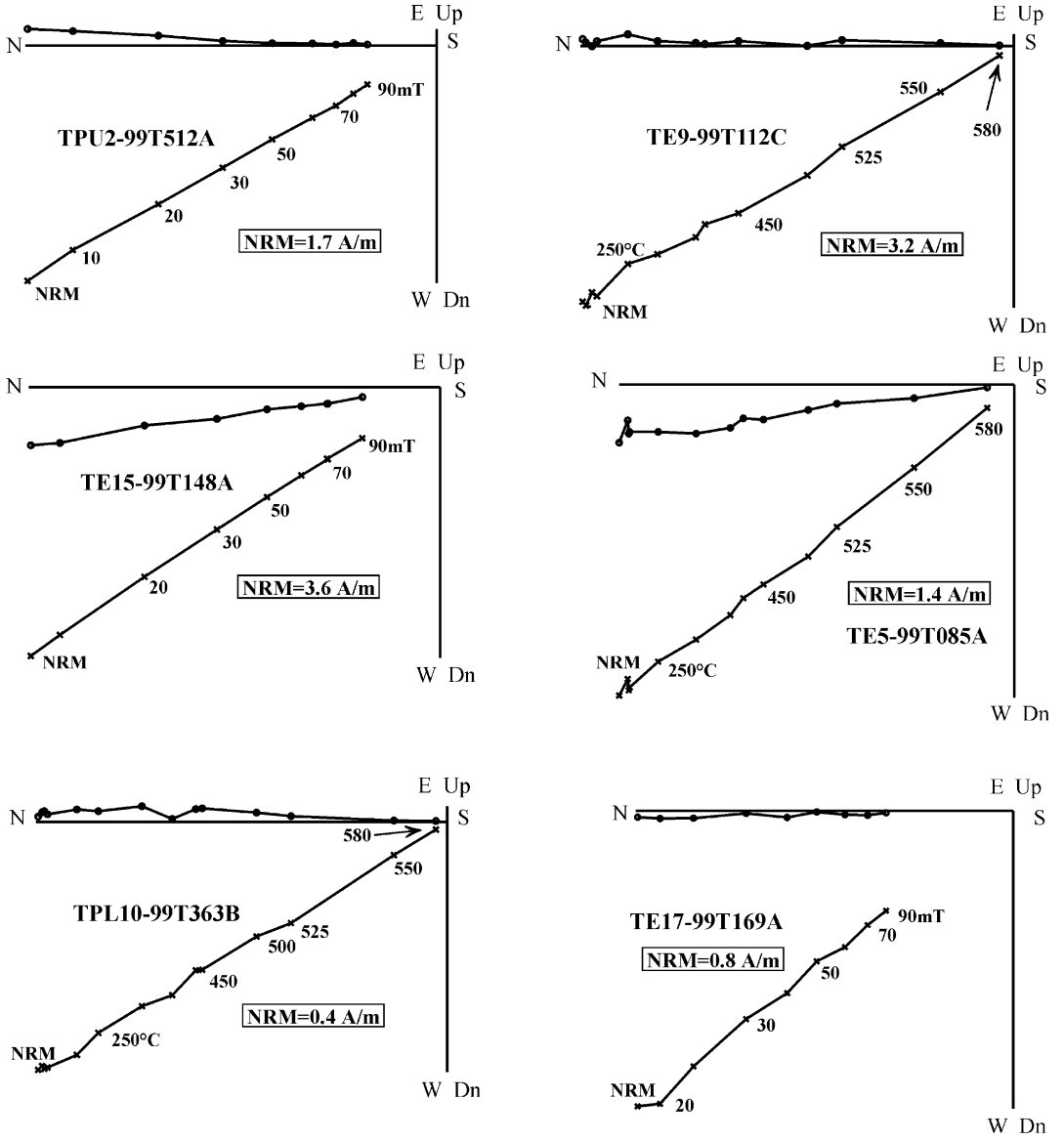


Fig. 7. Orthogonal vector plots of stepwise thermal or AF demagnetisation of representative samples (stratigraphic coordinates). The numbers refer either to the temperatures (°C) or to peak AFs (mT): (●) projections into the horizontal plane; (×) projections into the vertical plane.

not selected for Thellier and Thellier (1959) paleointensity experiments. Paleodirections, however, most probably remain unaltered by maghemitisation. Most of the experimental studies indicate that chemical remanent magnetisation due to maghemitisation are recorded in the same directions as the original thermoremanence (Heider and Dunlop, 1987; Nishitani and Kono, 1989; Goguitchaichvili et al., 2000a).

3.3. Hysteresis experiments

Hysteresis measurements at room temperature were performed on all studied samples in fields up to 1.4 T using the AGFM 'Micromag' apparatus of the paleomagnetic laboratory at Mexico City. The saturation remanent magnetisation (J_{rs}), the saturation magnetisation (J_s) and coercitive force (H_c) were calculated after correction for the paramagnetic contribution. The coercivity of remanence (H_{cr}) was determined by applying a progressively increasing backfield after saturation. Some typical hysteresis plots are shown in Fig. 6. The curves are symmetrical in almost all cases (sample 99T161). Near the origin, no potbellied and wasp-waisted behaviours (Tauxe et al., 1996) were detected, which probably reflects very restricted ranges of the opaque mineral coercivities. Only exception is site TPL1 (Fig. 6, sample 99T304), which probably reflects the presence of ferrimagnetic phases with different coercivities. Most probably this behaviour is due to presence of dominantly single domain and superparamagnetic grains (Goguitchaichvili and Prévot, 2000). Alternatively, similar behaviour may be observed if some (titano)hematites co-exist together with (titano)magnetites. For other samples, 'small' pseudo-single-domain grains seem to be responsible for remanence judging from hysteresis parameters values (Day et al., 1977). Let us note that if some superparamagnetic fraction exists in these samples, the measured coercive force would be somewhat lower and the saturation magnetisation larger.

3.4. Stepwise demagnetisation

The remanent magnetisation of five to eight samples from each lava flow (Table 1) were measured with a JR-5a and JR-6 (AGICO Ltd.) spinner magnetometers (sensitivity $\sim 10^{-9}$ Am^2) at paleomagnetic laboratory of National University of Mexico (UNAM). Measurements were recorded after stabilisation of the remanence in this magnetometer. Both alternating field (AF) demagnetisation using a laboratory made AF demagnetiser and stepwise thermal demagnetisation up to 580–650 °C using a non-inductive Schonstedt furnace were carried out. During thermal demagnetisation, the low-field susceptibility at room

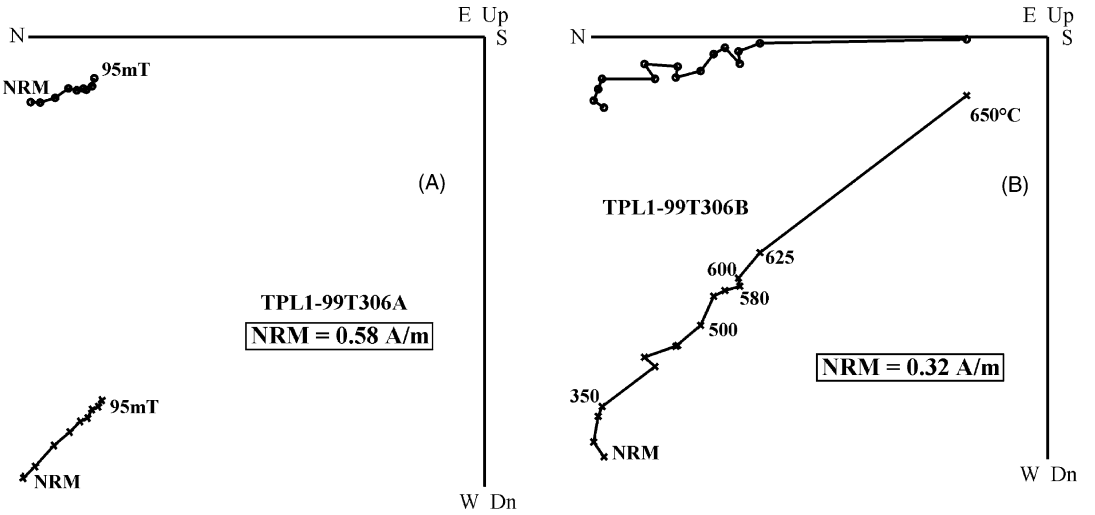


Fig. 8. Orthogonal vector plots of stepwise thermal demagnetisation of a sample containing (titano)hematite or a mixture of (titano)magnetite and (titano)hematite (see also text). Same notations as in Fig. 7.

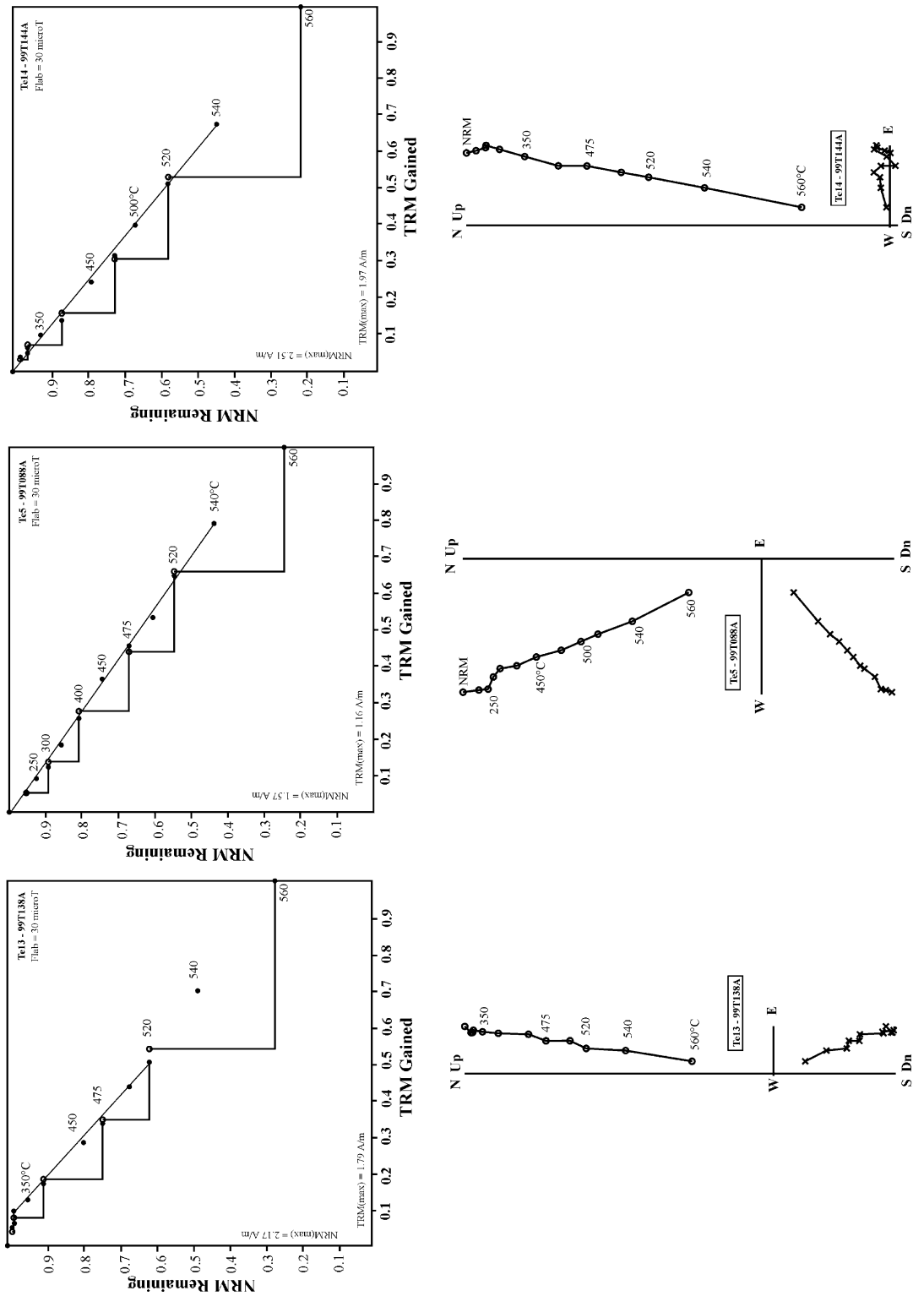


Fig. 9. The representative NRM–TRM plots and associated orthogonal diagrams from Tepic samples. In the orthogonal diagrams we used the same notations as in Fig. 7.

temperature was measured after each step with a Bartington susceptibility meter.

In almost all cases only one stable paleomagnetic component could be recognized (Fig. 7). Minor secondary components were easily removed applying 200 °C or 10 mT. In most cases, the greater part of remanent magnetisation was removed at temperatures between 500 and 580 °C, which indicates low-Ti

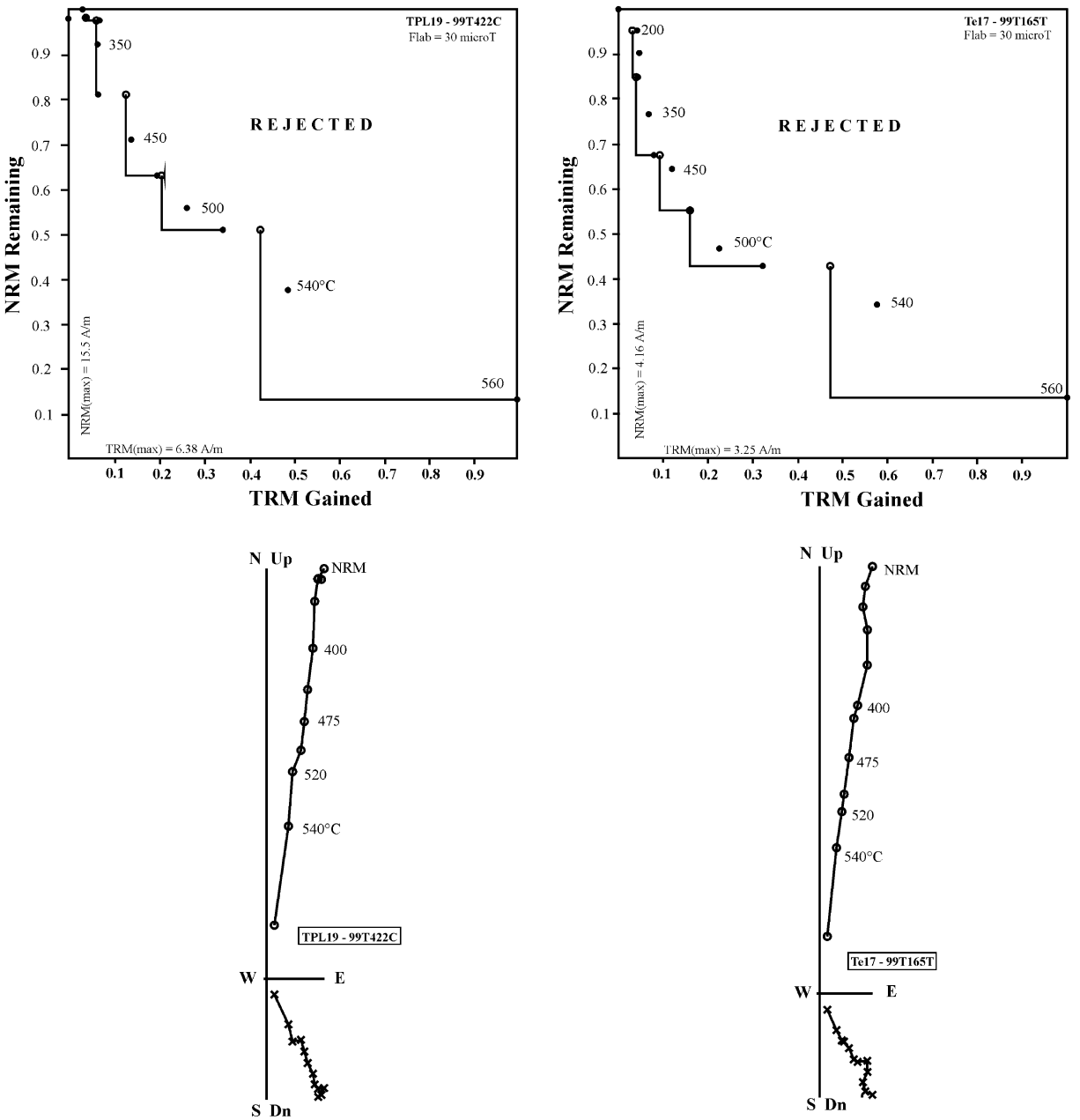


Fig. 10. Examples of rejected samples, showing a typical concave-up behaviour during Thellier paleointensity experiments (see also text). Same notations as in Fig. 9.

titanomagnetites to be responsible for magnetisation. Median destructive fields (MDF) range mostly from 40 to 50 mT suggesting 'small' pseudo-single-domain grains as remanent magnetisation carriers (Dunlop and Özdemir, 1997). One site (Fig. 8, site TPL1) yields evidence for (titano)hematite or a mixture of (titano)magnetite and (titano)hematite. Remanent magnetisation is not completely destroyed at 650 °C and application of a 95 mT alternating peak field only removes 25% of the initial remanence.

3.5. Paleointensity experiments

Paleointensity experiments were carried out in vacuum better than 10^{-2} mbar using a MDT80 paleointensity oven. The remanence measurements were made in a field-free environment in the paleomagnetic laboratory of the UNAM. Temperature reproducibility between two heatings to the same temperature was generally within 3 °C up to 450 °C and 2 °C above. The intensity of the laboratory field was 30 μT and was held at a precision better than 0.15 μT.

Table 2
Paleointensity results from Tepic volcanic units

Site	Sample	n	T_{min}–T_{max}	f	g	q	$F_E \pm \sigma(F_E)$	VDM	$F_E \pm$ S.D.	VDMe
TE1	99T060C	7	200–500	0.64	0.82	12.8	31.2 ± 1.3	6.77	29.9 ± 1.8	6.5 ± 0.4
	99T061A	10	200–560	0.82	0.85	20.8	32.1 ± 1.2	6.97		
	99T062C	7	200–500	0.81	0.81	19.7	28.9 ± 0.9	6.27		
	99T063D	7	200–500	0.82	0.83	20.4	27.6 ± 0.9	5.99		
	99T064C	7	200–500	0.78	0.82	17.6	29.9 ± 1.1	6.49		
TE5	99T083A	11	20–540	0.68	0.87	31.7	28.2 ± 0.5	6.84	30.5 ± 2.5	6.4 ± 0.8
	99T084C	10	200–520	0.44	0.85	11.9	32.5 ± 1.2	7.01		
	99T086A	11	20–540	0.67	0.86	31.2	29.0 ± 0.5	6.73		
	99T087A	10	200–540	0.68	0.85	23.1	33.9 ± 0.8	4.99		
	99T088B	12	20–540	0.69	0.87	36.7	29.3 ± 0.5	6.32		
TE9	99T109A	11	20–540	0.52	0.89	9.9	43.7 ± 1.8	10.1	42.8 ± 1.3	9.8 ± 0.3
	99T111A	9	200–520	0.46	0.87	7.3	41.8 ± 2.3	9.56		
TE13	99T134B	8	250–520	0.45	0.84	5.3	41.6 ± 3.6	9.12	36.2 ± 6.2	7.9 ± 1.4
	99T135A	8	250–520	0.38	0.83	5.8	37.5 ± 2.1	8.22		
	99T136B	9	200–520	0.41	0.85	5.8	42.5 ± 2.6	9.31		
	99T138A	7	250–520	0.51	0.83	11.1	30.7 ± 1.2	6.73		
	99T139C	10	200–540	0.69	0.81	13.5	28.8 ± 1.2	6.31		
TE14	99T144A	11	20–540	0.68	0.84	17.7	31.1 ± 0.9	7.03	32.8 ± 1.7	7.4 ± 0.4
	99T142B	9	250–540	0.66	0.83	7.9	34.4 ± 2.4	7.78		
	99T146C	8	300–540	0.52	0.81	17.4	33.0 ± 1.8	7.46		
TPL5	99T329B	8	350–520	0.41	0.85	6.3	27.4 ± 1.5	5.83	27.8 ± 0.9	5.9 ± 0.2
	99T330C	8	300–500	0.40	0.87	7.6	29.1 ± 1.3	6.19		
	99T332C	9	250–500	0.41	0.88	7.0	26.9 ± 1.4	5.72		
	99T333C	9	250–500	0.39	0.88	7.1	27.7 ± 1.3	5.89		
TPL7	99T343B	9	300–520	0.46	0.81	7.5	41.3 ± 2.1	9.29	42.0 ± 7.9	9.5 ± 1.8
	99T345C	9	300–520	0.51	0.78	5.6	50.3 ± 3.5	11.3		
	99T346C	10	200–540	0.47	0.87	16.1	34.5 ± 0.9	7.76		
TPU4	99T520B	8	250–500	0.48	0.80	8.2	28.8 ± 1.4	6.16	30.5 ± 1.7	6.5 ± 0.3
	99T523C	8	250–500	0.47	0.81	6.6	29.8 ± 1.7	6.38		
	99T525D	7	300–500	0.48	0.80	8.7	30.7 ± 1.4	6.57		
	99T526C	8	300–520	0.53	0.82	7.3	32.3 ± 1.9	6.91		

Here n is the number of NRM–TRM points used for palaeointensity determination; T_{min}–T_{max} the temperature interval used; f, g and q the fraction of extrapolated NRM used, the gap factor and the quality factor (Coe et al., 1978), respectively; F_E the paleointensity estimate for an individual specimen; $\sigma(F_E)$ its standard error; F_E the average paleointensity of individual lava flow; the plus and minus sign corresponding to standard deviation; and VDM and VDMe are individual and average virtual dipole moments, respectively.

We used the Coe version of the Thellier method (Coe, 1967) with sliding NRM–TRM checks (Prévot et al., 1985). At each temperature step the samples were heated twice: in zero field for the first heating and in presence of field for the second heating. The pTRM checks were performed after every second step throughout the whole experiment.

Following paleodirectional and rock-magnetic results, 64 samples belonging to 14 cooling units, yielded stable, one component magnetisations with blocking temperatures compatible to a Ti-poor titanomagnetite phase, low viscosity index (lower than 4%) and nearly reversible *k–T* curves were pre-selected for Thellier paleointensity experiments. Paleointensity data are reported on the classical Arai-Nagata (Nagata et al., 1963) plot on Figs. 9 and 10 and results are given in Table 2. We accepted only determinations that fulfil the following criteria: (1) determination obtained from at least seven NRM–TRM points corresponding to a NRM fraction larger than one-third (Table 2); (2) quality factor (Coe et al., 1978) of about five or more; and (3) positive 'pTRM' checks, i.e. the deviation of 'pTRM' checks were less than 15%. Directions of NRM remaining at each step obtained from the paleointensity experiments are reasonably linear and point to the origin. No deviation of remaining NRM directions towards the direction of the applied laboratory field was observed.

Finally, only 31 samples from eight individual basaltic lava flows yielded acceptable paleointensity estimates. For these samples the NRM fraction *f* used for determination ranges between 0.38 and 0.82 and the quality factor *q* from 5.3 to 36.7 being generally greater than 5. For remained samples, the main reason for failure of Thellier paleointensity experiments was a typical 'concave-up' behaviour (Dunlop and Özdemir, 1997). An important loss of NRM was observed without any noticeable TRM acquisition (Fig. 10). This phenomenon can be due to irreversible variations of coercive force (Kosterov and Prévot, 1998) at low temperature and can be interpreted as transformation from a single-domain 'metastable' state to polydomain state which results in a large NRM lost without any correlated TRM acquisition during the subsequent cooling.

4. Results and discussion

A characteristic magnetisation direction was determined by the least squares method (Kirschvink, 1980) with 7–11 points being taken in the principal

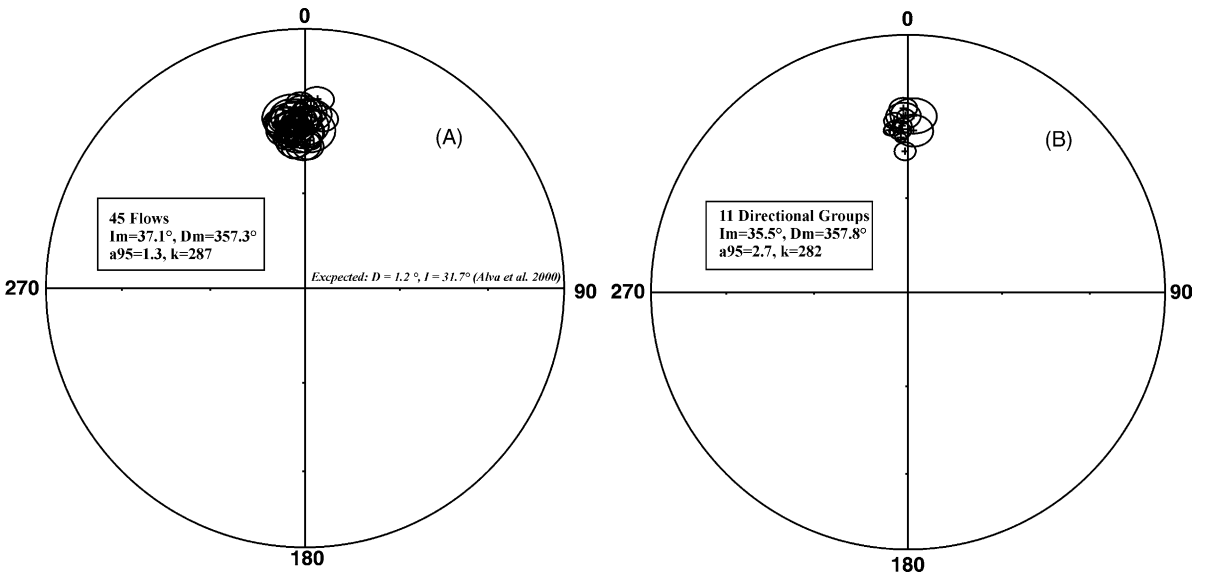


Fig. 11. Equal area projections of the flow-mean characteristic paleodirections (A) and DG (see text for explanation) mean paleodirections (B) of Nayarit volcanics.

component analysis. The obtained directions of each cooling unit were averaged and the statistical parameters calculated assuming a Fisherian distribution. These average unit directions are very well defined (Table 1, Fig. 11): all α_{95} are below 6.8°. This points to a very high quality paleomagnetic data.

All 45 lava flows show normal magnetic polarity. With this result, combined with single Ar–Ar datum of 8.93 million years, a direct correlation with the geomagnetic polarity time scale (GPTS) can be made. This correlation points to chron C4Ar.1n (Cande and Kent, 1995) as emplacement time interval of Tepic volcanic succession. Thus, it may be speculated that the Tepic volcanic units have been emplaced during a very short time span of 0.08 million years or even less.

The mean paleodirection obtained in this study is $I = 37.1°$, $D = 357.3°$, $k = 287$, $\alpha_{95} = 1.3°$, $N = 45$, which corresponds to the mean paleomagnetic pole position $P_{lat} = 87.3°$, $P_{long} = 324.8°$, $A_{95} = 1.1°$, $K = 366$. It is apparent that each lava flow does not record an independent measurement of the geomagnetic field. For this reason, samples from several flows that record the same field direction have been averaged together, and the resulting direction are given as a 'directional group' (DG) number in Table 3. DGs were defined following Mankinen et al.'s (1985) criteria (see also Prévot et al., 1985 and Camps and Prévot, 1996). In general, if the flow-mean paleodirections of two or more flows showed no systematic trends and their ovals of 95% confidence overlapped, they were considered to record the same field directions. In this way we defined 11 DGs (Table 3). Each DG contain 1–12 lava flows. The mean paleodirection according to DGs is $I = 35.5°$, $D = 357.8°$, $k = 282$, $\alpha_{95} = 2.7°$, $N = 282$. This directions are in good agreement with the expected paleodirections for middle Miocene time (Fig. 11), as derived from reference poles given by Besse and Courtillot (1991) for North America (see also Alva-Valdivia et al., 2000). This probably suggests that no major tectonic deformation occurred in the studied area since about 9 million years.

The classical formula $S_F^2 = S_T^2 - S_W^2/n$ was used for estimating paleosecular variation in this study. Here, S_T is the total angular dispersion $S_T = [(1/N-1)\sum_{i=1}^{N}\delta i^2]^{1/2}$ (Cox, 1969), N the number of DGs used in calculation and δi is the angular distance of the ith virtual geomagnetic pole (VGP) from the axial dipole; S_W represents the within site dispersion, n the average number of samples per unit. McFadden et al. (1991) report a VGP scatter equal to 18.1 for the 20° latitude bands between 5 and 22.5 million years. We obtained $S_F = 4.9$ (upper and lower limits: 7.6–3.6), which is significantly lower (Fig. 12). The evidence of relatively low secular variation from central Mexico was already reported by Goguitchaichvili et al. (2001, Fig. 12). We tried to compare VGP scatter obtained in this study to the dispersion obtained from the other Miocene volcanic sites, preferentially from American continent. There are only few studies directly available. In most cases, evidence for

Table 3
Summary of paleodirectional and paleointensity results for each DGs (see also text for the definition of DGs)

DG	N	n	Inc	Dec	α_{95}	k	N'	n'	$F_E \pm$ S.D.	VDMe
11	6	39	37.3	357.2	3.1	471	2	9	30.1 ± 1.6	6.5 ± 0.3
10	3	18	44.4	358.8	3.1	1686				
9	3	22	37.2	1.9	5.6	481	1	5	30.5 ± 2.5	6.4 ± 0.8
8	3	18	31.9	1.8	6.4	377	1	2	42.8 ± 1.3	9.8 ± 0.3
7	1	6	31.9	359.2	2.3	158				
6	7	43	36.3	354.9	2.5	601	2	8	35.0 ± 5.1	7.7 ± 1.1
5	1	6	33.5	355.3	2.8	177				
4	12	79	39.7	357.3	1.6	767	2	7	33.9 ± 8.9	7.4 ± 2.1
3	3	17	32	358.5	4.8	660				
2	5	29	36.8	353.6	1.8	1831				
1	1	7	29.2	358.5	3.8	252				

N and n are the numbers of independent lava flows and samples, respectively, to define a DG; Dec the declination; Inc the inclination; k and α_{95} the precision parameter and radius of confidence cone, respectively; N' and n' the numbers of flows and samples, respectively, which yielded reliable paleointensity estimates within each DG; F_E the average paleointensity; and VDMe is the average virtual dipole moment for each DGs.

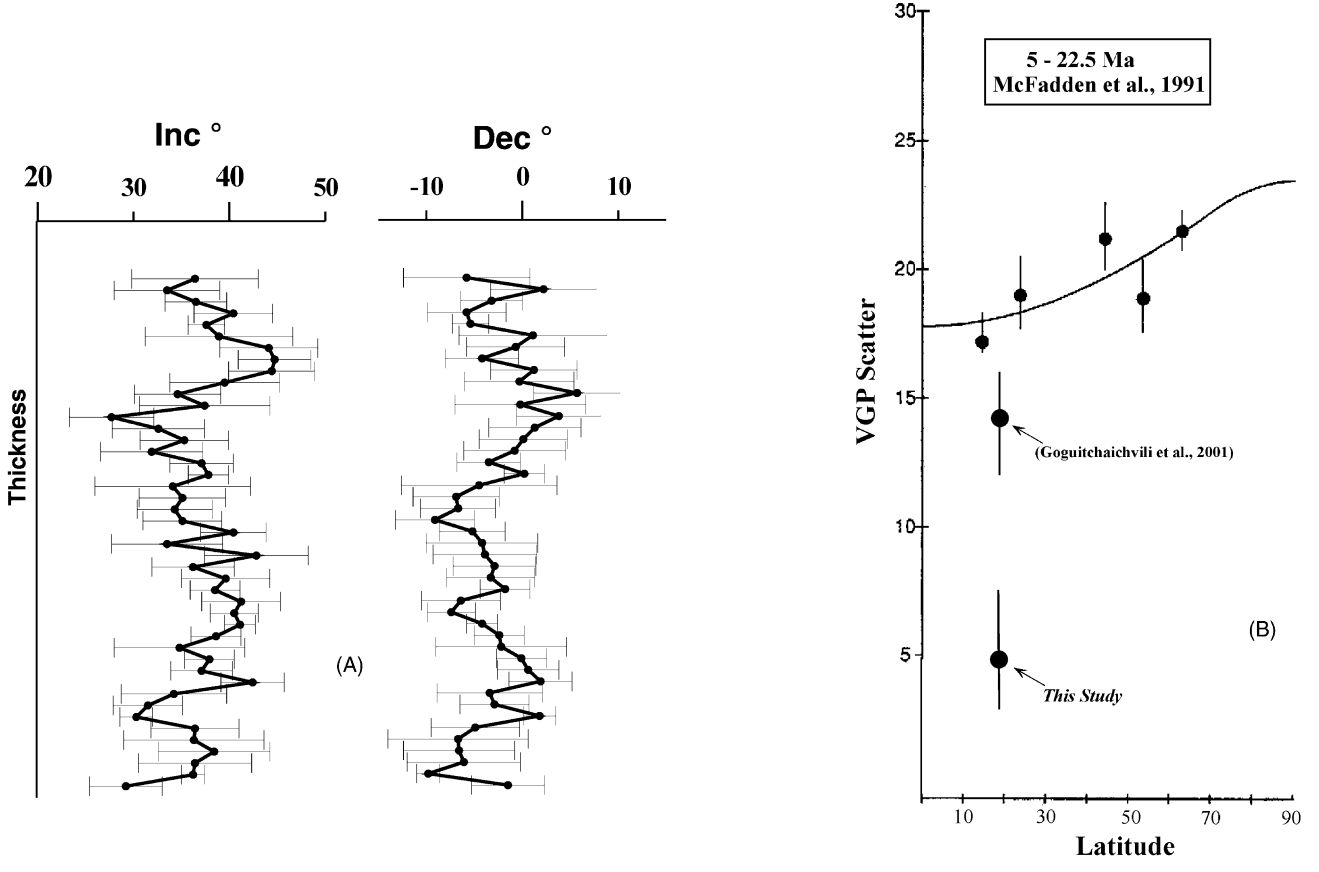


Fig. 12. (A) Flow mean, inclination and declination for each flow of Tepic section against elevation. (B) Paleosecular variation of lavas (PSVL) between 5 and 22.5 million years (retrieved from McFadden et al., 1991).

Table 4
Paleomagnetic and paleosecular variation parameters from Miocene volcanic formations

Locality	Reference	Site	Age (million year)	N	Dec°	Inc°	k	α_{95}°	S_T	S_U	S_L
Chile	GO-2000	Farellones	∼17	27	20.3	55.5	15.2	7.4	21	25.8	17.7
Peru	TS-1987	Titicaca	5–6	5	350.3	23.4	22.3	16.6	17.2	28.4	12.3
Argentina	SO-1996	Puna	8.9	16	358.2	31.2	21.9	8.1	17.9	23.5	14.5
Indian Ocean	HE-1997	Kerguelen	∼21	48	6.9	64.3	21	4.5	16.7	19.4	14.7
Siria	RO-1986	S. Damasco	19.5	24	357.2	35.1	22.1	6.5	17.4	21.6	14.6
Oregon	PR-1985	Steens Mtn.	16.25	80	0.1	65.5	26.2	3.2	15.9	17.8	14.3
Central Mexico	GO-2001	Rio Grande	8–9	37	354.6	31.1	124	2.1	11.1	13.2	9.6
Western Mexico	This study	Tepic	∼9	11 (DG)	357.8	35.5	282	2.7	4.8	7.5	3.6

N is the number of studied units; Dec and Inc the site mean magnetic declination and inclination, respectively; k and α_{95} the precision parameter and 95% confidence cone of the Fisher statistics, respectively; S_T the total between flow dispersion of average flow remanence direction (see text); and S_U and S_L are the 95% confidence limits for S_T after Cox (1969).

local tectonic rotations are reported and thus they cannot be correctly used for paleosecular variation study. The VGP changing from flow to flow found at Steens Mountain (Mankinen et al., 1985) volcanic succession (discarding intermediate polarity units) agree quite well with McFadden et al.'s (1991) model yielding VGP scatter equal to 18.9. The same calculation on Steens Mountain DGs (Prévot et al., 1985) provide basically the same value. Up to now, there are very few studies reporting paleomagnetic results in Miocene volcanic rocks for the South American continent (Goguitchaichvili et al., 2000a). Most studies belong to the Andean chain and it is thus difficult to separate local tectonic rotation from true paleosecular variation (Roperch et al., 1996). For this reason we chose to calculate the angular distance from the site mean direction and then S_T. Analyses of Table 4 shows that the S_T value of the Tepic volcanic succession is abnormally low compared to other regions.

It may be speculated that the extremely low paleosecular variation documented in this study supports the hypothesis that the central Pacific region might have been a region of low non-dipole geomagnetic field in the Miocene as suggested by Doell and Cox (1971) for the Brunhes chron (see also Tanaka and Kono, 1991; Mankinen and Champion, 1993; Elmaleh et al., 2001). Alternatively, our results may indicate a relatively high volcanic eruption rate and thus very short emplacement time for the lava flows.

Thirty-one samples from eight individual flows yielded acceptable paleointensity estimates. The site mean paleointensities are ranging from 27.8 ± 0.9 to $42.0 \pm 7.9\,\mu$T. The virtual dipole moments (VDM) range from 5.9 to 9.5×10^{22} Am^2. This correspond to mean value of $7.6 \pm 1.4 \times 10^{22}$ Am^2, which is higher than average VDM value for late Miocene (Table 5, selected data only, see Goguitchaichvili et al., 2000b). In total, our data suggest the existence of relatively high geomagnetic field strength undergoing some relatively low fluctuations. These results support Love's (2000) suggestion about inverse relationship between secular variation and local field strength as result of electomagnetic coupling between the solid inner core and liquid outer core, with the inner core tending to stabilise core convection, and hence the field, when intensity is high. It is worth noting that some

Table 5
Average virtual dipole moment (10^{22} Am^2) for various regions during the Miocene (from selected Thellier palaeointensity data)

Region	Age (million year)	N	VDM (10^{22} Am^2)	S.D.	Reference
Nayarit	∼9	8	7.6	1.4	This study
Jalisco	8.1–10.3	4	7.1	1.2	Goguitchaichvili et al. (2000a,b)
Central France	8.6–9.7	3	3.9	2.7	Riisager (1999)
Oceanic Basalts	6.4–10.3	2	4.8	4.2	Juàrez et al. (1998)
North America	∼15.5	10	6.9	2.4	Prévot et al. (1985)

N refers to the number of cooling units used to calculate the average VDM.

variation of absolute intensity was detected within the same DG (Tables 2 and 3). Somewhat similar results were found by Prévot et al. (1985) studying Steens Mountain volcanic succession and Goguitchaichvili et al. (2001) studying a single volcanic sequence from southern Caucasus. Similar to these authors we believe that the absolute paleointensity variation of individual flows from the same DG may be a real geomagnetic phenomenon rather than an experimental noise, indicating that the intensity of the geomagnetic field varies faster than its direction.

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